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LAB 4: PROBABILITY DISTRIBUTIONS

⇒ "RANDOM VARIABLE" – VALUE RESULTS FROM MEASUREMENT OF A QUANTITY SUBJECT TO RANDOM VARIATIONS

Types – Discrete
– Continuous

"INDEPENDENT & IDENTICALLY DISTRIBUTED" (IID)
– VARIABLES DRAWN FROM SAME DISTRIBUTION, THAT ARE INDEPENDENT

↳ IIDRV if $p(x,y) = p(x)p(y)$

⇒ PROBABILITY & PROBABILITY DENSITY

IN A CONTINUOUS DISTRIBUTION, THE PROBABILITY OF MEASURING ANY <u>PARTICULAR</u> VALUE IS ZERO! THE BEST WE CAN DO IS DEFINE A "PROBABILITY DENSITY FUNCTION" (PDF) SUCH THAT...

$$\int_{x_1}^{x_2} f(x)\,dx = \text{Probability of measuring } x_1 < X < x_2$$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

⇒ Descriptive Statistics (location, scale, shape of PDF)

- Mean (expectation value) $\mu = E(x) = \int_{-\infty}^{\infty} x\, f(x)\, dx$

- Variance $(\text{std. dev.})^2$ $\sigma^2 = V = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx$

- Skewness (asymmetry) $\Sigma' = \int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right)^3 f(x)\, dx$

- Kurtosis (heavy-tailedness) $K = \underbrace{\underbrace{\int_{-\infty}^{\infty} \left(\frac{x-\mu}{\sigma}\right)^4 f(x)\, dx}_{\text{excess kurtosis}} - 3}_{= \phi \text{ for Normal/Gaussian}}$

Collectively "moments" of distribution $f(x)$

- Absolute deviation about d $\delta = \int_{-\infty}^{\infty} |x-d|\, f(x)\, dx$

- Mode (most probable value) $\left.\frac{df(x)}{dx}\right|_{x=x_m} = \phi$

⇒ ESTIMATES OF DESCRIPTIVE STATISTICS FROM DATA

MEAN $\mu = \int_{-\infty}^{\infty} x f(x)\, dx \rightarrow \bar{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$

STDDEV $\sigma = \sqrt{\int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx} \rightarrow S = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2}$

"BESSEL'S CORRECTION"
(lose a degree of freedom WHEN DERIVING $\bar{x}$ from data!)

UNCERTAINTIES $\sigma_{\bar{x}} = \frac{S}{\sqrt{N}}$

$$\sigma_S = \frac{S}{\sqrt{2(N-1)}}$$

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LAB 4: EXPONENTIAL DISTRIBUTION

$f(x) = \frac{1}{\beta} e^{-x/\beta}$ ← PYTHON CONVENTION!

$\{$ALT $f(x) = \beta e^{-\beta x}\}$

$\int_0^{\infty} \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{\beta} \left[-\beta e^{-x/\beta}\right]_0^{\infty} = 1$ ✓

<NOTEBOOK: rv_gallery.ipynb>